File 42: Nationwide Permit 27 PCN Technical Template

***Regulatory Permitting & Case Studies Library Template Type**: Nationwide Permit 27 Pre-Construction Notification (PCN) **Federal Authority**: Clean Water Act Section 404 / USACE Savannah District **Supervising Board**: Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)*

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1. Application Instructions

This document serves as a high-fidelity template for preparing a **Clean Water Act Section 404 Nationwide Permit 27 (NWP 27)** Pre-Construction Notification (PCN) package.
When preparing a submittal for a new high-gradient stream restoration tract in North Georgia, copy this technical narrative, fill in the HUC-8 basin-specific geomorphic data, and attach the corresponding AutoCAD plan sheets and HEC-RAS floodway modeling reports.

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2. USACE Form ENG 4345 Technical Narrative Template

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U.S. ARMY CORPS OF ENGINEERS - NATIONWIDE PERMIT 27 ATTACHMENT
PRE-CONSTRUCTION NOTIFICATION (PCN) TECHNICAL NARRATIVE
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APPLICANT: Blue Ridge Stream Restoration & Mitigation LLC
MANAGING DIRECTOR: Hunter Morris
BOARD MEMBER: Hadi Irvani (University of Virginia - UVA / HBS)
PROJECT NAME: Anderson Creek Fluvial Restoration Project (Roya's Cabin Reach)
LOCATION: Fannin County, Georgia (Latitude: 34.8456° N, Longitude: 84.3412° W)
WATERSHED: Upper Savannah River Basin (HUC 03060102)
STREAM REACH LENGTH: 1,500 Linear Feet

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1. PROJECT PURPOSE AND NEED
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The primary ecological objective of this project is to restore natural fluvial geomorphic
stability, improve aquatic wild trout habitats, and establish a native riparian forest buf
fer along a severely degraded 1,500-linear foot reach of Anderson Creek.

Historically, this reach was cleared of native vegetation and subjected to unmanaged cattl
e grazing. These land practices resulted in:
A. Extreme bank erosion and active slope failures, contributing over 85 tons of fine sedim
ent per year downstream.
B. Vertical channel incisement (Rosgen Type G/F channel) with loss of floodplain access.
C. Smothering of historic cobble substrate with fine silts, wiping out benthic macroinvert
ebrates.
D. Direct solar warming, raising water temperatures above the 65 degree F Wild Brook Trout
 spawning thermal threshold.

There is a critical need to stabilize these bank failures and regrade the reach into a sel
f-scouring pool-riffle channel to restore coldwater freshwater habitats.

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2. DETAILED DESCRIPTION OF PROPOSED ECOLOGICAL RESTORATION
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To achieve permanent geomorphic and biological lift, the Developer will implement Natural
Channel Design (NCD) and geomorphic regrading:

A. BANK REGRASTING AND FLOODPLAIN RE-MEANDERING:
The existing steep, eroding banks (slopes up to 1:0.5) will be regraded to a stable 1:2 or
 1:3 slope. A bankfull floodplain bench (15 feet wide) will be excavated to allow peak flo
odways to expand safely onto the valley flats, reducing bank shear stresses.

B. INSTREAM STRUCTURES (cedar logs and oak root-wads):
- J-Hook Rock Vanes (15 structures): Built along outer curves to deflect high-velocity cur
rents toward the center of the channel, preventing bank erosion and scouring deep pool hab
itats.
- Double Log Cross-Vanes (5 structures): Built at riffle-pool transitions to provide verti
cal grade control, protect the stream bed from incision, and create plunge pools.
- Cedar Toe-Wood (450 linear feet): Placed below the baseflow waterline along regraded cur
ves to provide structural bank toe stabilization and physical overhead cover for trout.

C. RIPARIAN BUFFER ESTABLISHMENT:
We will plant a minimum 100-foot wide native riparian forest buffer on both stream banks.
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The buffer will feature a high density of mature Rhododendron maximum transplants, tag ald
ers, eastern hemlocks, and river birches (target density >= 320 stems/acre).

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3. ENVIRONMENTAL IMPACTS & AVOIDANCE AND MINIMIZATION MEASURES
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The project is strictly designed to restore ecological habitats and will result in zero ne
t loss of aquatic resources. Temporary construction impacts have been minimized as follows
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A. DRY CHANNEL BYPASS (PUMP-AROUND SEQUENCING):
All physical grading and instream structure placement will occur "in the dry" using a temp
orary bypass channel:
1. A temporary gravel-bag sand dandy coffer dam will be constructed upstream of the active
 reach.
2. High-volume pumps will divert all stream flow around the work zone through flexible pip
elines, discharging water downstream through a sediment filter bag before re-entering the
main channel.
3. Heavy equipment (GPS-guided excavators) will operate entirely within the dry channel be
d, preventing turbid sediment releases.

B. IMPACT SUMMARY:
- Temporary Streambed Disturbance: 1,500 Linear Feet (0.68 Acres).
- Permanent Fill: 0.00 Acres (all J-hooks and log vanes utilize native rocks and wood, res
ulting in net geomorphic expansion, not fill).
- Wetland Impacts: 0.00 Acres.

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4. COMPLIANCE STATIONS AND MATHEMATICAL CALCULATIONS
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We have processed geomorphic surveys to ensure design stability:
- Entrenchment Ratio (ER): Pre-restoration ER = 1.15 (entrenched); Post-restoration design
 ER = 2.45 (slightly entrenched, stable).
- Width-to-Depth Ratio (WDR): Pre-restoration WDR = 26.5 (overwidened, silt-choking); Post
-restoration design WDR = 14.8 (stable transport geometry).
- Sinuosity (K): Pre-restoration K = 1.08; Post-restoration design K = 1.35.

HEC-RAS floodway modeling has been completed for the 100-year peak-flow storm event. The w
ater surface profiles demonstrate a complete 'No-Rise' (Delta WSEL <= 0.00 feet) across al
l cross-sections.
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3. Mandatory Engineering Attachments Checklist

Ensure the following plan sheets and certifications are compiled and appended to this PCN technical package:

- [ ] **Sheet G-01: Cover & Index Sheet**: Formatted with Hunter Morris's credentials and Hadi Irvani's UVA strategic board approval seal.
- [ ] **Sheet C-01: Plan-view Alignment Sheet**: AutoCAD Civil 3D layout showing bankfull meanders, radius of curvature, and coordinate points for all rock J-hooks and log cross-vanes.
- [ ] **Sheet C-02: Geomorphic Cross-Sections**: Detailed comparative drawings of pre- and post-restoration cross-sections showing Bankfull Stage, Floodprone Stage ($2 \cdot D_{max}$), and bank slopes.

- [ ] **Sheet D-01: Structural Details**: Detailed, geotechnically sound structural drawings of coir-wrapped soil lifts (GRSL), toe-wood layering, and J-hook rock anchoring specifications.
- [ ] **Sheet E-01: Erosion & Sedimentation Control**: ES&PC plan showing pump-around details, coffer dam coordinates, sediment filter bags, and floating turbidity curtains.
- [ ] **FEMA No-Rise Certification**: Hydrologic engineering statement signed and sealed by a Georgia Registered Professional Engineer (PE).